Code No: 135AF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August/September - 2024
DESIGN AND ANALYSIS OF ALGORITHMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	In what way a time complexity differs from space complexity.	[2]
b)	How can we measure an algorithm's running time?	[3]
c)	What is a set? List the operations that can be performed on it.	[2]
d)	Define spanning tree and minimum spanning tree.	[3]
e)	What is the importance of knapsack algorithm in our daily life?	[2]
f)	What is greedy strategy? Explain.	[3]
g)	What is reliability design?	[2]
h)	What are the applications of dynamic programming?	[3]
i)	What is NP-hard problem?	[2]
j)	What is meant by non-deterministic algorithm?	[3]

PART – B

(50 Marks)

2.a) What are the different mathematical notations used for algorithm analysis?
b) Distinguish between Merge sort and quick sort. Give suitable examples. [5+5]

OR

3.a) Explain about Recursive Binary search algorithm with suitable examples.
b) Write Randomized algorithm of Quicksort. [5+5]

4.a) Write and explain the algorithm of Bi connected components with an example.
b) Explain about union and find algorithms. [5+5]

OR

5.a) Discuss about the graph traversal and coloring with an example.
b) Write an algorithm to find the strongly connected components in a diagraph. [5+5]

6.a) State the Job – Sequencing with deadlines problem.
b) Find an optimal sequence to the n = 5 Jobs where
Profits (P1, P2, P3, P4, P5) = (20, 15, 10, 5, 1) and
Deadlines (d1, d2, d3, d4, d5) = (2, 2, 1, 3, 3) are given. [5+5]

OR

7.a) Write the pseudo code for Dijkstra's algorithm for single source shortest path problem.
b) Write an algorithm of Prim's minimum cost spanning tree. [5+5]

8. How Matrix – chain Multiplication problem can be solved using dynamic programming? Explain with suitable example. [10]

OR

9.a) Draw an Optimal Binary Search Tree for n=4 identifiers (a1,a2,a3,a4) = (do, if, read, while) P(1:4)=(3,3,1,1) and Q(0:4)=(2,3,1,1,1).

b) Describe the travelling sales person problem. [5+5]

10.a) Explain Cook's theorem and its significance.

b) Discuss in detail about FIFO Branch and Bound solutions. [5+5]

OR

11.a) Explain the classes of NP Hard and NP-Complete.

b) Show that the Hamiltonian cycle problem on directed graphs is NP-complete. [5+5]

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